Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this assignment is to examine a Google Chrome History database for digital forensic evidence. It aims to identify browser activities related to suspected drug advertising, communication, and payment.
The investigation involves examining URLs, visit records, timestamps, and downloaded files. It also requires reconstructing the order of browser activities using SQL queries.

2. Lab Report Structure

The report begins with the case details, executive summary, and evidence acquisition process. It explains the tools, database tables, and methods used to examine the Chrome History file. The findings describe relevant browser records, navigation activities, timestamps, and downloads. A chronological timeline and evidence log are included to support the investigation.

2.1. Title Page

- Lab Title: Case Study 2: Reconstructing Chrome Web History
- Student Name: Olanrewaju Wale Mustapha Oladimeji.
- Student ID: 2025/FWSD/11320.
- Course Name: SBT-DF204: Computer Forensic Case Study
- Instructor Name: Aminu Idris
- Date of Submission: 10/10/2026
- Version:

2.2. Executive Summary

This investigation examines a supplied Chrome History SQLite database for relevant browser activity. The analysis uses SQL queries to identify URLs, visit records, navigation patterns, and downloads. The recorded timestamps are converted to UTC to help establish the sequence of events.
Relevant records are assessed for possible links to suspected advertising, communication, and

payment activities.
The conclusion will depend on the actual database findings and will acknowledge the limitations of browser history evidence.

2.3. Lab Objectives

The first objective is to preserve the supplied History database and verify its integrity using SHA-256 hashing. The second objective is to identify relevant database tables and fields using SQLite queries. The third objective is to examine browser visits, navigation transitions, timestamps, and download records. The fourth objective is to reconstruct a chronological timeline and document supporting evidence.
The final objective is to evaluate the findings carefully while identifying limitations and avoiding unsupported conclusions.

2.4. Tools and Resources Used

Kali Linux is used as the main environment for conducting the digital forensic investigation.
SQLite is used to examine the Chrome History database and retrieve relevant records.
The Linux terminal is used to copy evidence, check file details, and calculate SHA-256 hashes.
SQL queries are used to analyse URLs, visits, timestamps, navigation transitions, and downloads.
The supplied History database, saved query results, screenshots, timeline, and evidence log provide the main investigation resources.

2.5. Methodology

This individual case study asks you to examine a Chrome History SQLite database and reconstruct an evidence-based timeline of suspected drug trafficking activity. You will identify meaningful browser records, convert timestamps, examine navigation transitions, and connect observations across postings, communications, payment references, and downloads. Your report must distinguish what the browser history directly records from what you infer about the user or alleged activity.

Case scenario

Investigators received a Google Chrome History file from a device associated with a suspect. The case hypothesis states that the browser may have been used to advertise controlled drugs, communicate with an interested buyer, exchange payment information, verify a cryptocurrency transaction, and retain proof of payment. Your task is to assess the available history evidence

and construct a defensible timeline. The case requires forensic interpretation of a supplied database, not interaction with live websites, accounts, messages, or wallets.

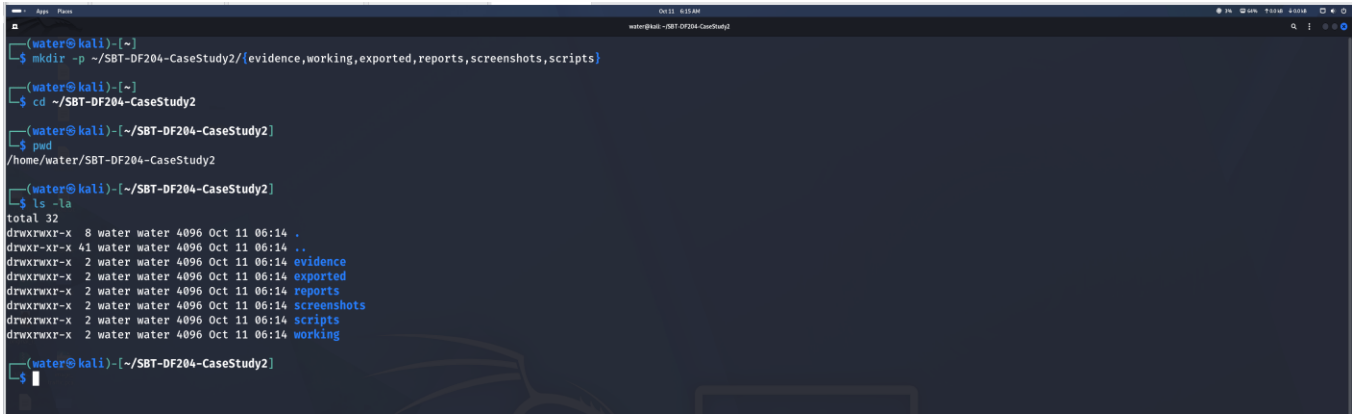
Case materials

The supplied case file is a Chrome History SQLite database named **History**. The teaching source shown in the case materials is: added in study material section

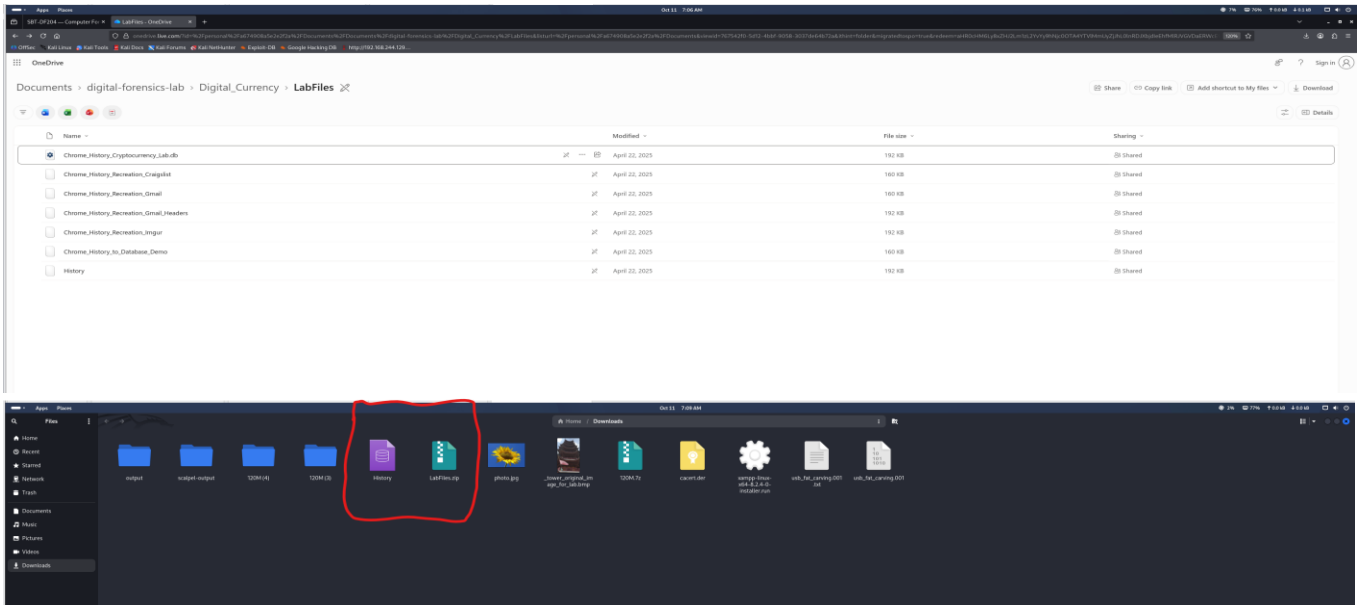
Create a working copy before opening the database. Record the source, original file name, file size, and SHA-256 value. Use a SQLite examination tool such as DB Browser for SQLite or sqlite3. Preserve screenshots and SQL output that support your findings.

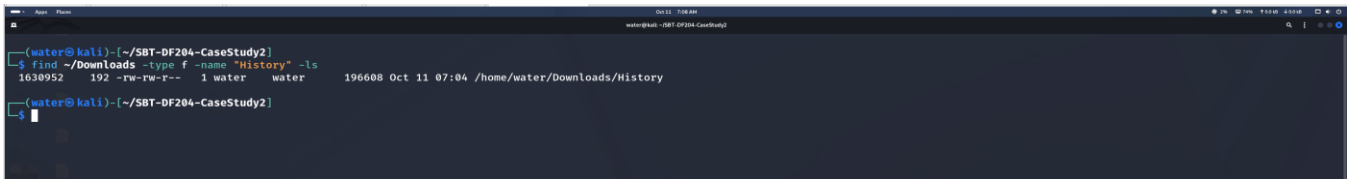
Creating Case Study 2 Folder

mkdir -p ~/SBT-DF204-CaseStudy2/{evidence,working,exported,reports,screenshots,scripts}



Downloading the Chrome History SQlite database name History





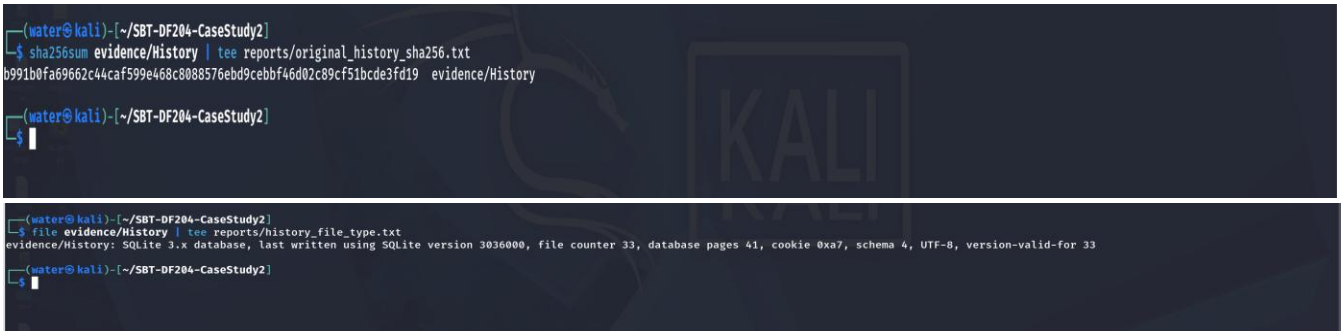
Record the source, original file name, file size, and SHA-256 value



SHA-256 Value

sha256sum evidence/History | tee reports/original_history_sha256.txt

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History



Creating Forensic Working Copy, confirming the SHA-256 and comparing the evidence history to working copy history

cp --preserve=mode,timestamps evidence/History working/History_working

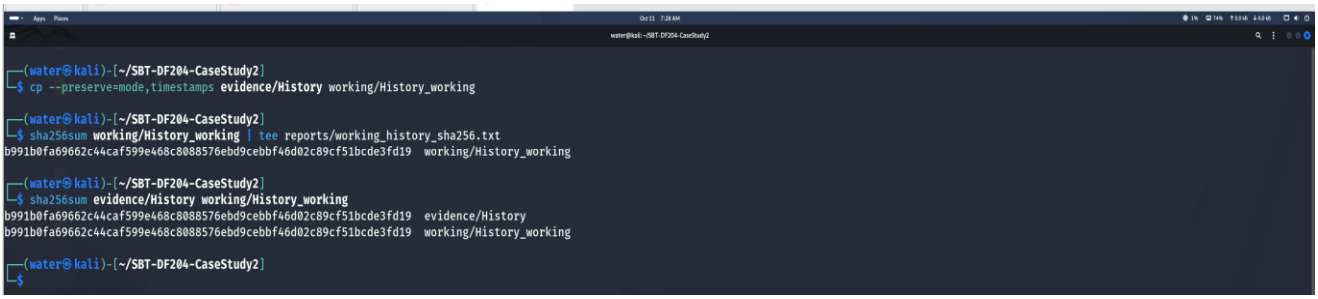
sha256sum working/History_working | tee reports/working_history_sha256.txt

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History_working

sha256sum evidence/History working/History_working

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History

b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History_working

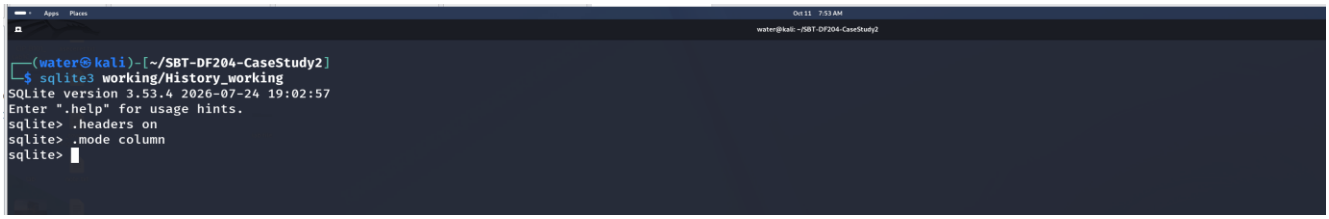


Checking SQLite 3 Version

sqlite3 –version



Opening the Chrome History database



Identifying the tables

SELECT name

FROM sqlite_master

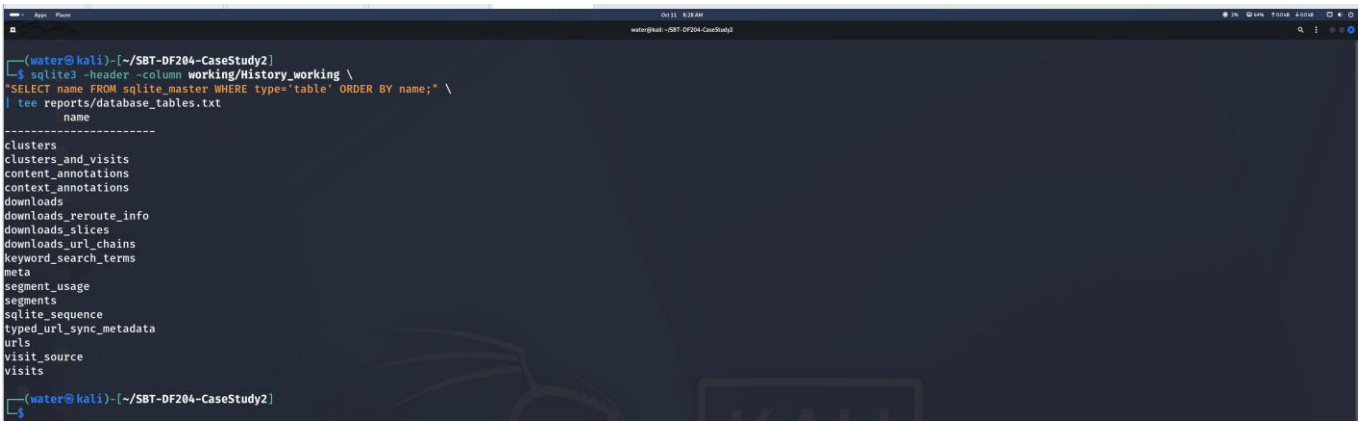
WHERE type = 'table'

ORDER BY name; or use the command below

sqlite3 -header -column working/History_working \

"SELECT name FROM sqlite_master WHERE type='table' ORDER BY name;" \

| tee reports/database_tables.txt



```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 working/History_working
SQLite version 3.53.4 2026-07-24 19:02:57
Enter ".help" for usage hints.
sqlite> .headers on
sqlite> .mode column
sqlite> SELECT name
FROM sqlite_master
WHERE type = 'table'
ORDER BY name;
        name
----------------------
clusters
clusters_and_visits
content_annotations
context_annotations
downloads
downloads_reroute_info
downloads_slices
downloads_url_chains
keyword_search_terms
meta
segment_usage
segments
sqlite_sequence
typed_url_sync_metadata
urls
visit_source
visits
sqlite>
```

Inspecting the database structure

PRAGMA table_info(urls);

PRAGMA table_info(visits);

PRAGMA table_info(downloads);

PRAGMA table_info(downloads_url_chains);

```
sqlite> PRAGMA table_info(urls);
PRAGMA table_info(visits);
PRAGMA table_info(downloads);
PRAGMA table_info(downloads_url_chains);
cid       name           type      notnull  dflt_value  pk
---  --------------  ----------  -------  ----------  --
  0  id              INTEGER           0              1
  1  url             LONGVARCHAR       0              0
  2  title           LONGVARCHAR       0              0
  3  visit_count     INTEGER           1  0           0
  4  typed_count     INTEGER           1  0           0
  5  last_visit_time  INTEGER          1              0
  6  hidden          INTEGER           1  0           0
cid             name                          type     notnull  dflt_value  pk
---  -------------------------------  -------  -------  ----------  --
  0  id                               INTEGER        0              1
  1  url                              INTEGER        1              0
  2  visit_time                       INTEGER        1              0
  3  from_visit                       INTEGER        0              0
  4  transition                       INTEGER        1  0           0
  5  segment_id                       INTEGER        0              0
  6  visit_duration                   INTEGER        1  0           0
  7  incremented_omnibox_typed_score  BOOLEAN        1  FALSE       0
  8  opener_visit                     INTEGER        0              0
cid          name                type        notnull  dflt_value  pk
---  ----------------------  ----------  -------  ----------  --
  0  id                      INTEGER           0              1
  1  guid                    VARCHAR           1              0
  2  current_path            LONGVARCHAR       1              0
  3  target_path             LONGVARCHAR       1              0
  4  start_time              INTEGER           1              0
  5  received_bytes          INTEGER           1              0
  6  total_bytes             INTEGER           1              0
  7  state                   INTEGER           1              0
  8  danger_type             INTEGER           1              0
  9  interrupt_reason        INTEGER           1              0
 10  hash                    BLOB              1              0
 11  end_time                INTEGER           1              0
 12  opened                  INTEGER           1              0
 13  last_access_time        INTEGER           1              0
 14  transient               INTEGER           1              0
 15  referrer                VARCHAR           1              0
 16  site_url                VARCHAR           1              0
 17  tab_url                 VARCHAR           1              0
 18  tab_referrer_url        VARCHAR           1              0
 19  http_method             VARCHAR           1              0
 20  by_ext_id               VARCHAR           1              0
 21  by_ext_name             VARCHAR           1              0
 22  etag                    VARCHAR           1              0
 23  last_modified           VARCHAR           1              0
 24  mime_type               VARCHAR(255)      1              0
 25  original_mime_type      VARCHAR(255)      1              0
 26  embedder_download_data  VARCHAR           1  ''          0
cid     name         type      notnull  dflt_value  pk
---  ----------  ----------  -------  ----------  --
  0  id          INTEGER           1              1
  1  chain_index  INTEGER          1              2
  2  url         LONGVARCHAR       1              0
sqlite>
```

Or

sqlite3 -header -column working/History_working \"PRAGMA table_info(urls);" \

| tee reports/schema_urls.txt

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working \
"PRAGMA table_info(urls);" \
| tee reports/schema_urls.txt
cid  name           type         notnull  dflt_value  pk
---  -------------  -----------  -------  ----------  --
0    id             INTEGER      0                    1
1    url            LONGVARCHAR  0                    0
2    title          LONGVARCHAR  0                    0
3    visit_count    INTEGER      1        0           0
4    typed_count    INTEGER      1        0           0
5    last_visit_time  INTEGER    1                    0
6    hidden         INTEGER      1        0           0

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
```

sqlite3 -header -column working/History_working \"PRAGMA table_info(visits);" \
| tee reports/schema_visits.txt

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working \
"PRAGMA table_info(visits);" \
| tee reports/schema_visits.txt
cid  name                          type     notnull  dflt_value  pk
---  ----------------------------  -------  -------  ----------  --
0    id                            INTEGER  0                    1
1    url                           INTEGER  1                    0
2    visit_time                    INTEGER  1                    0
3    from_visit                    INTEGER  0                    0
4    transition                    INTEGER  1        0           0
5    segment_id                    INTEGER  0                    0
6    visit_duration                INTEGER  1        0           0
7    incremented_omnibox_typed_score  BOOLEAN  1     FALSE       0
8    opener_visit                  INTEGER  0                    0

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
```

sqlite3 -header -column working/History_working \"PRAGMA table_info(downloads);" \
| tee reports/schema_downloads.txt

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working \
"PRAGMA table_info(downloads);" \
| tee reports/schema_downloads.txt
cid  name                 type         notnull  dflt_value  pk
---  -------------------  -----------  -------  ----------  --
0    id                   INTEGER      0                    1
1    guid                 VARCHAR      1                    0
2    current_path         LONGVARCHAR  1                    0
3    target_path          LONGVARCHAR  1                    0
4    start_time           INTEGER      1                    0
5    received_bytes       INTEGER      1                    0
6    total_bytes          INTEGER      1                    0
7    state                INTEGER      1                    0
8    danger_type          INTEGER      1                    0
9    interrupt_reason     INTEGER      1                    0
10   hash                 BLOB         1                    0
11   end_time             INTEGER      1                    0
12   opened               INTEGER      1                    0
13   last_access_time     INTEGER      1                    0
14   transient            INTEGER      1                    0
15   referrer             VARCHAR      1                    0
16   site_url             VARCHAR      1                    0
17   tab_url              VARCHAR      1                    0
18   tab_referrer_url     VARCHAR      1                    0
19   http_method          VARCHAR      1                    0
20   by_ext_id            VARCHAR      1                    0
21   by_ext_name          VARCHAR      1                    0
22   etag                 VARCHAR      1                    0
23   last_modified        VARCHAR      1                    0
24   mime_type            VARCHAR(255)  1                   0
25   original_mime_type   VARCHAR(255)  1                   0
26   embedder_download_data  VARCHAR   1  ''                0

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
```

Checking the number of records

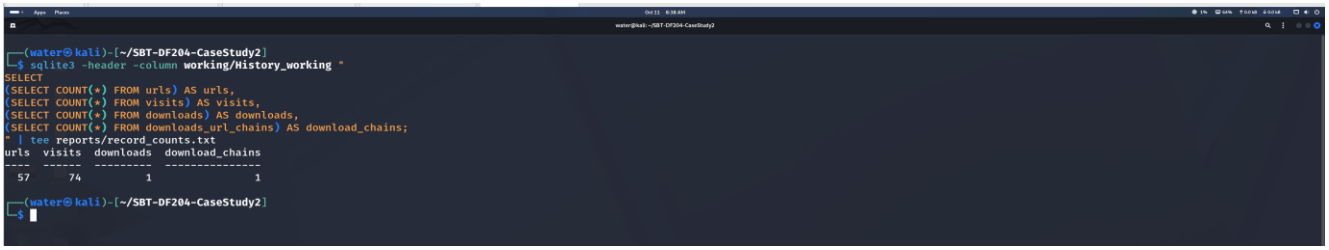
```
SELECT
   (SELECT COUNT(*) FROM urls) AS total_urls,
   (SELECT COUNT(*) FROM visits) AS total_visits,
```

(SELECT COUNT(*) FROM downloads) AS total_downloads;

Or

sqlite3 -header -column working/History_working "SELECT(SELECT COUNT(*) FROM urls) AS urls,(SELECT COUNT(*) FROM visits) AS visits,(SELECT COUNT(*) FROM downloads) AS downloads,(SELECT COUNT(*) FROM downloads_url_chains) AS download_chains;
" | tee reports/record_counts.txt



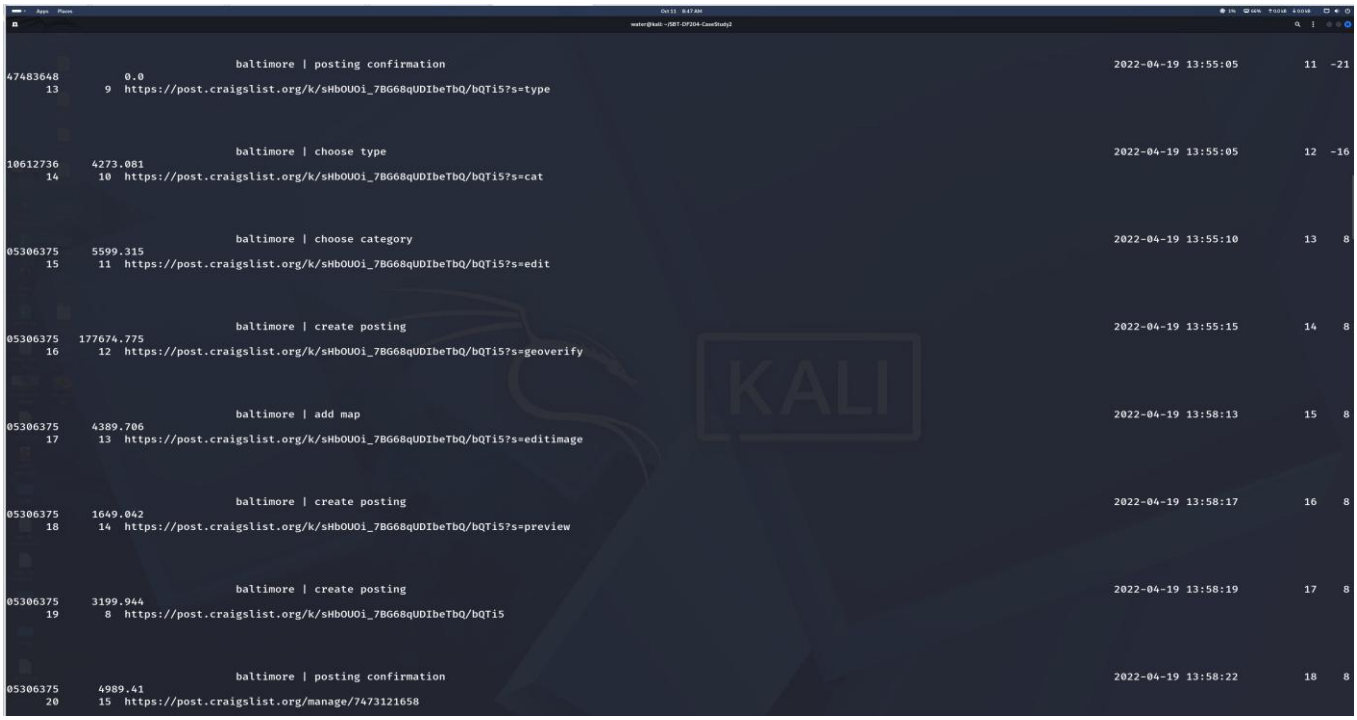


Build a visit timeline

sqlite3 -header -column working/History_working "
SELECT
  v.id AS visit_id,
  u.id AS url_id,
  u.url,
  u.title,
  datetime(
    v.visit_time/1000000 +
    strftime('%s','1601-01-01'),
    'unixepoch'
) AS visit_utc,
  v.from_visit,
  v.transition,
  v.visit_duration/1000.0 AS duration_ms
FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
" | tee reports/full_visit_timeline.txt

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working "
SELECT
    v.id AS visit_id,
    u.id AS url_id,
    u.url,
    u.title,
    datetime(
        v.visit_time/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS visit_utc,
    v.from_visit,
    v.transition,
    v.visit_duration/1000.0 AS duration_ms
FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
" | tee reports/full_visit_timeline.txt
visit_id  url_id
```

visit_id	url_id	url	title	visit_utc	from_visit	transition	duration_ms
1	1	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8	craigslist - Google Search	2022-04-19 13:54:18	0	805306373	0.0
2	1	https://www.google.com/search?q=craigslist&rlz=1C1ONGR_enUS989US989&oq=craigslist&aqs=chrome..69i57j46i131i199i433i465i512j0i402l2j0i512l3j0i131i433l2j0i512.2595j0j15&sourceid=chrome&ie=UTF-8	craigslist - Google Search	2022-04-19 13:54:19	0	805306368	727.413
3	2	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 13:54:19	2	268435456	0.0
4	3	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 13:54:19	3	-2147483648	0.0
5	4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 13:54:19	4	-1610612736	2465.46
6	5	https://accounts.craigslist.org/login/home	craigslist account	2022-04-19 13:54:22	5	268435456	0.0
7	6	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	craigslist - account log in	2022-04-19 13:54:22	6	-1610612736	17456.657
8	5	https://accounts.craigslist.org/login/home	craigslist account	2022-04-19 13:54:39	7	805306375	23439.248
9	2	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 13:55:03	8	268435456	0.0
10	4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events	2022-04-19 13:55:03	9	-1610612736	2664.177
11	7	https://post.craigslist.org/c/bal	baltimore | choose type	2022-04-19 13:55:05	10	268435456	0.0
12	8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5					



```
                                baltimore | posting confirmation                                                                2022-04-19 13:55:05          11  -21
47483648          0.0
      13       9  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type


                                baltimore | choose type                                                                       2022-04-19 13:55:05          12  -16
10612736     4273.081
      14      10  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat


                                baltimore | choose category                                                                   2022-04-19 13:55:10          13    8
05306375     5599.315
      15      11  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit

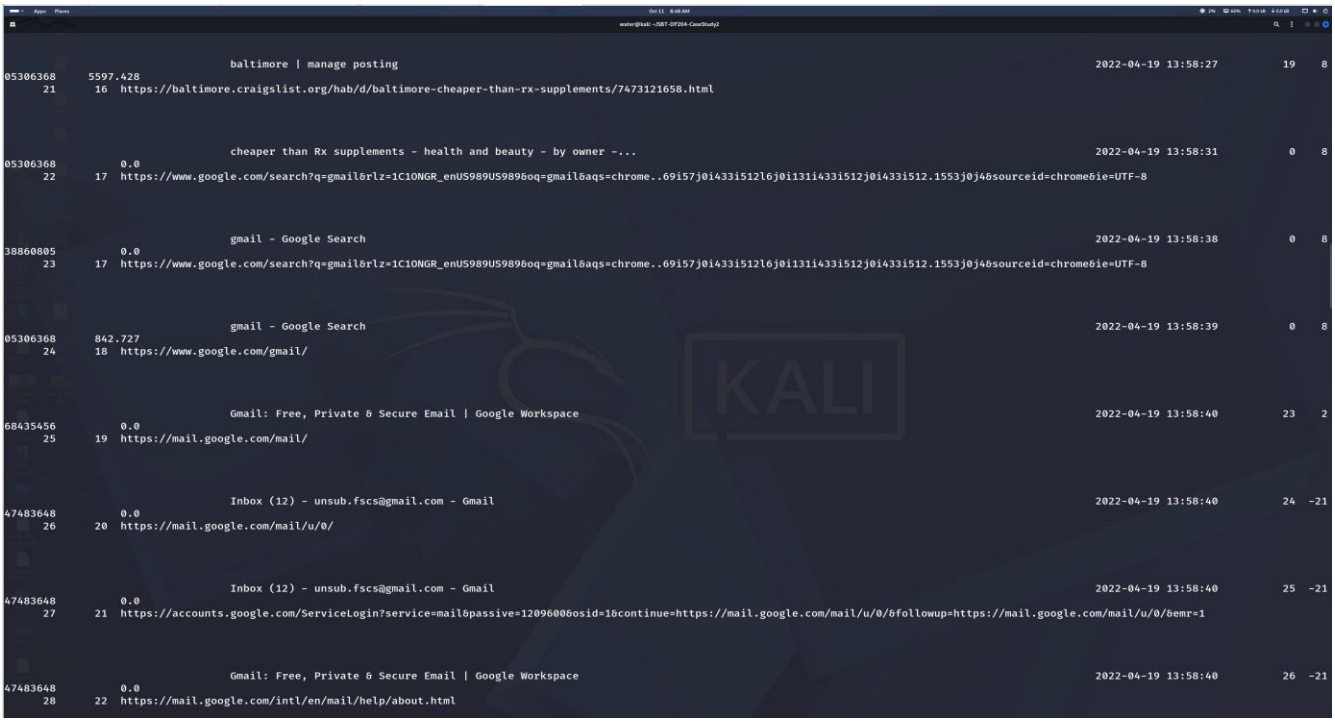

                                baltimore | create posting                                                                    2022-04-19 13:55:15          14    8
05306375   177674.775
      16      12  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify


                                baltimore | add map                                                                          2022-04-19 13:58:13          15    8
05306375     4389.706
      17      13  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage


                                baltimore | create posting                                                                   2022-04-19 13:58:17          16    8
05306375     1649.042
      18      14  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview


                                baltimore | create posting                                                                   2022-04-19 13:58:19          17    8
05306375     3199.944
      19       8  https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5


                                baltimore | posting confirmation                                                            2022-04-19 13:58:22          18    8
05306375      4989.41
      20      15  https://post.craigslist.org/manage/7473121658
```



```
                                baltimore | manage posting                                                                  2022-04-19 13:58:27          19    8
05306368     5597.428
      21      16  https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html


                                cheaper than Rx supplements - health and beauty - by owner -...                              2022-04-19 13:58:31           0    8
05306368          0.0
      22      17  https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chrome..69i57j0i433i512l6j0i131i433i512j0i433i512.1553j0j4&sourceid=chrome&ie=UTF-8


                                gmail - Google Search                                                                        2022-04-19 13:58:38           0    8
38860805          0.0
      23      17  https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chrome..69i57j0i433i512l6j0i131i433i512j0i433i512.1553j0j4&sourceid=chrome&ie=UTF-8

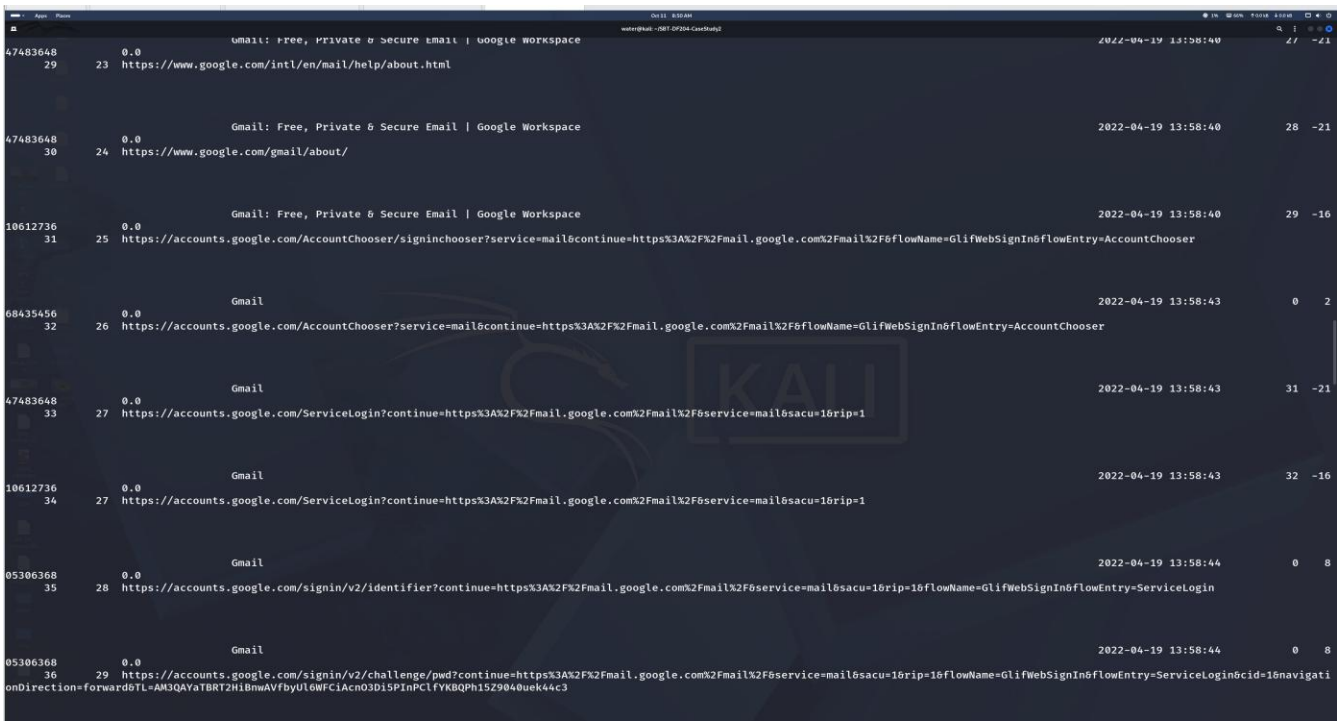

                                gmail - Google Search                                                                        2022-04-19 13:58:39           0    8
05306368      842.727
      24      18  https://www.google.com/gmail/


                                Gmail: Free, Private & Secure Email | Google Workspace                                       2022-04-19 13:58:40          23    2
68435456          0.0
      25      19  https://mail.google.com/mail/


                                Inbox (12) - unsub.fscs@gmail.com - Gmail                                                    2022-04-19 13:58:40          24  -21
47483648          0.0
      26      20  https://mail.google.com/mail/u/0/


                                Inbox (12) - unsub.fscs@gmail.com - Gmail                                                    2022-04-19 13:58:40          25  -21
47483648          0.0
      27      21  https://accounts.google.com/ServiceLogin?service=mail&passive=1209600&osid=1&continue=https://mail.google.com/mail/u/0/&followup=https://mail.google.com/mail/u/0/&emr=1


                                Gmail: Free, Private & Secure Email | Google Workspace                                       2022-04-19 13:58:40          26  -21
47483648          0.0
      28      22  https://mail.google.com/intl/en/mail/help/about.html
```



```
                                Gmail: Free, Private & Secure Email | Google Workspace                                                  2022-04-19 13:58:40          27  -21
47483648          0.0
      29      23  https://www.google.com/intl/en/mail/help/about.html



                                Gmail: Free, Private & Secure Email | Google Workspace                                                  2022-04-19 13:58:40          28  -21
47483648          0.0
      30      24  https://www.google.com/gmail/about/



                                Gmail: Free, Private & Secure Email | Google Workspace                                                  2022-04-19 13:58:40          29  -16
10612736          0.0
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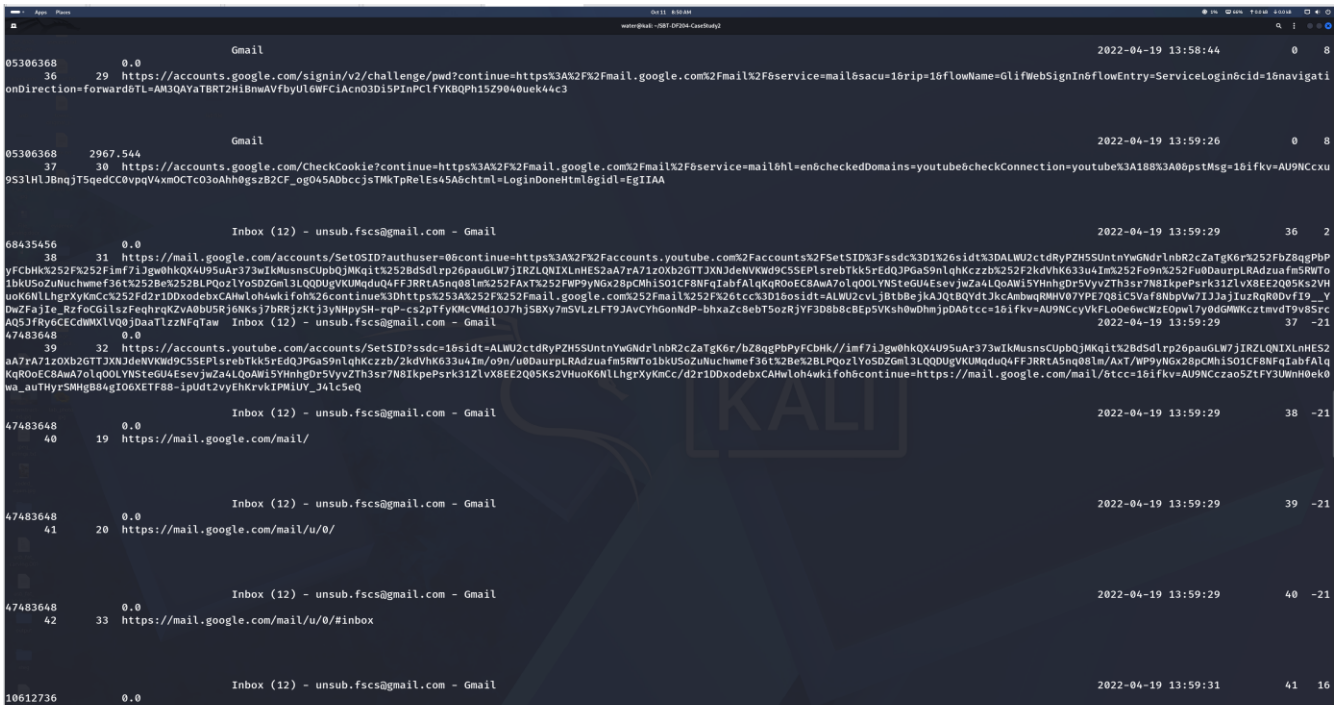
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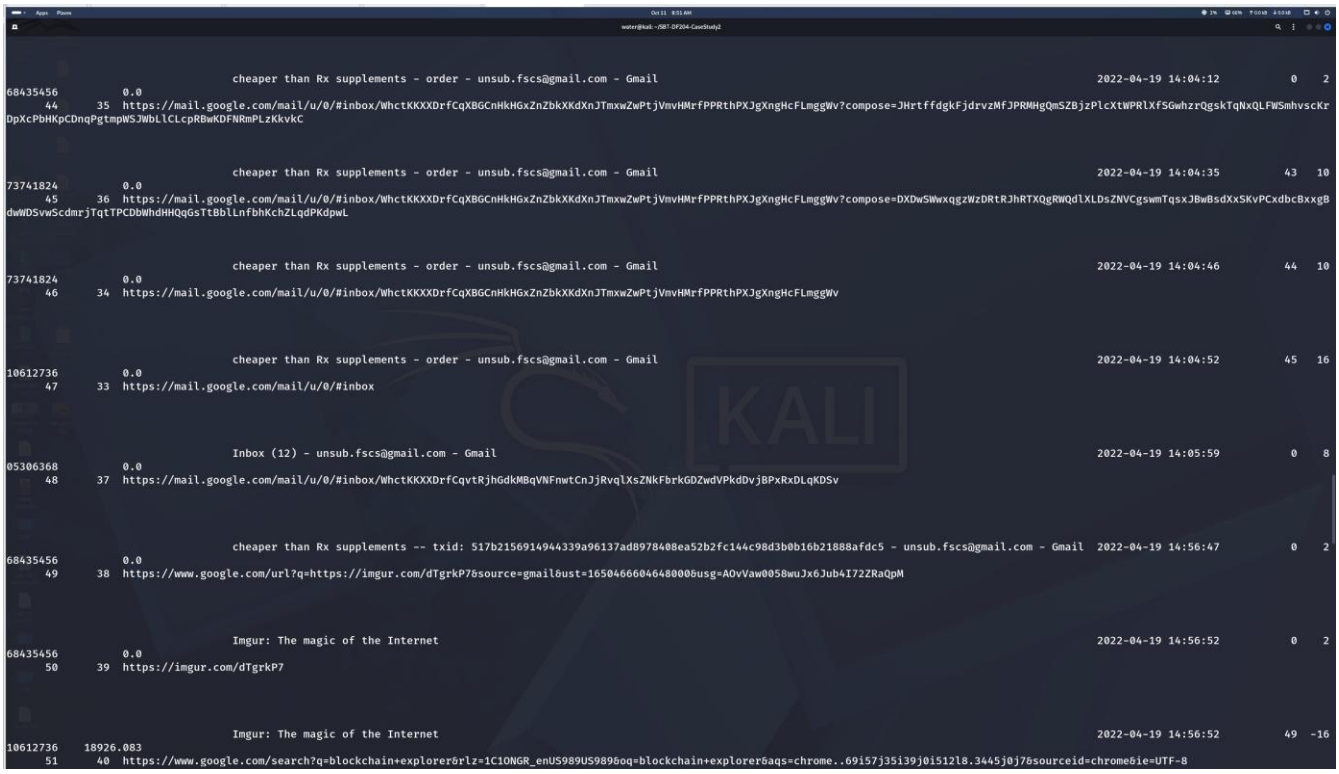
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73741824          0.0
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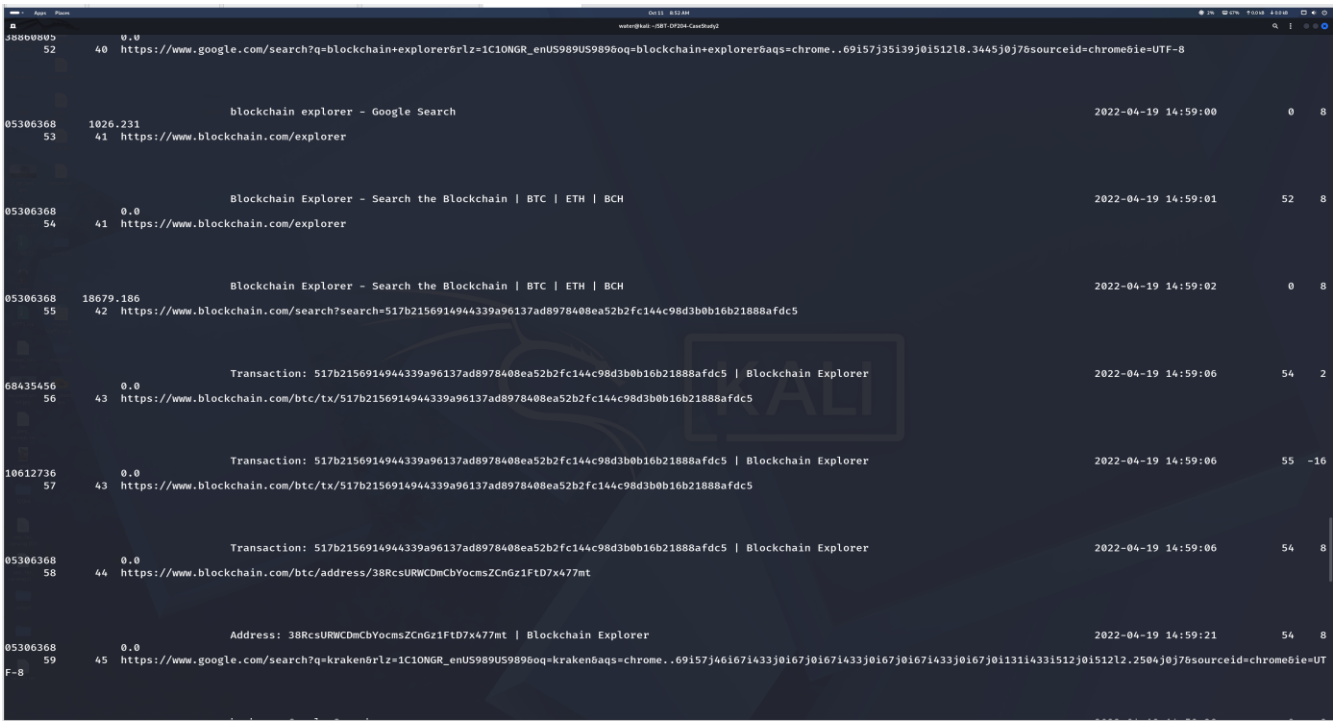

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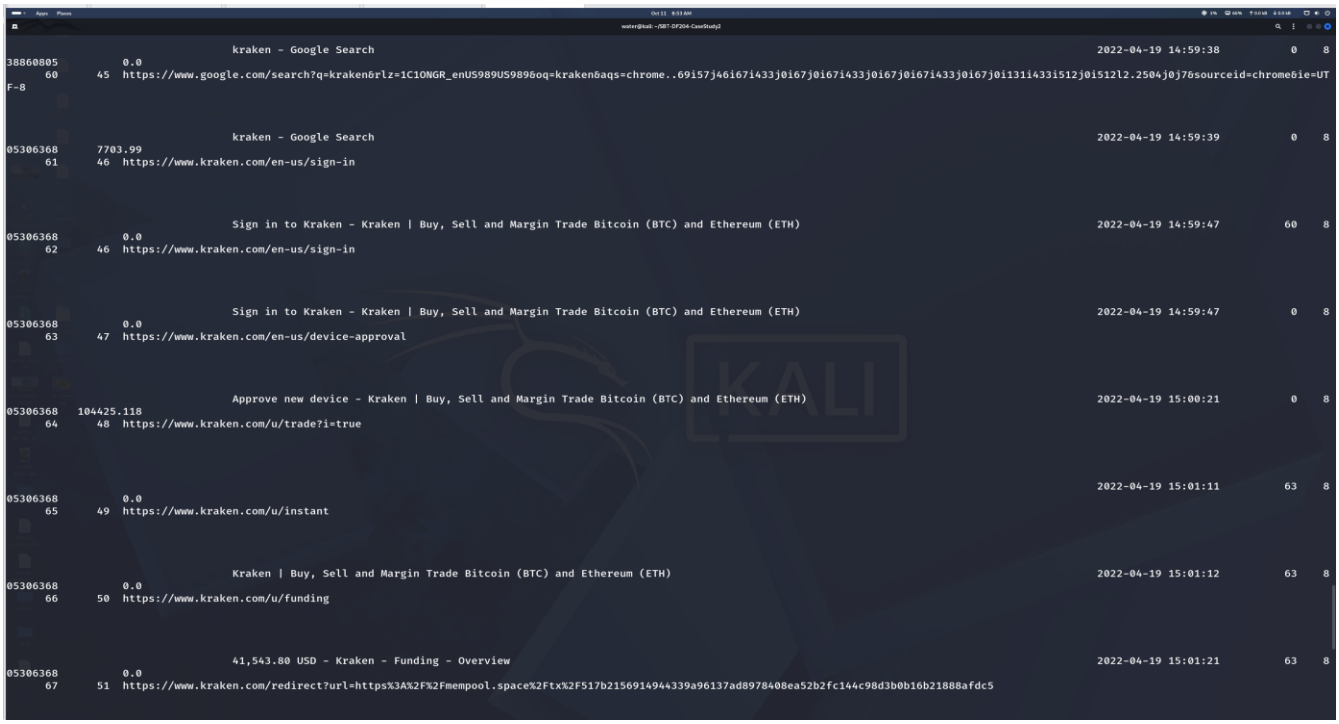

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05306368       7703.99
      61      46  https://www.kraken.com/en-us/sign-in



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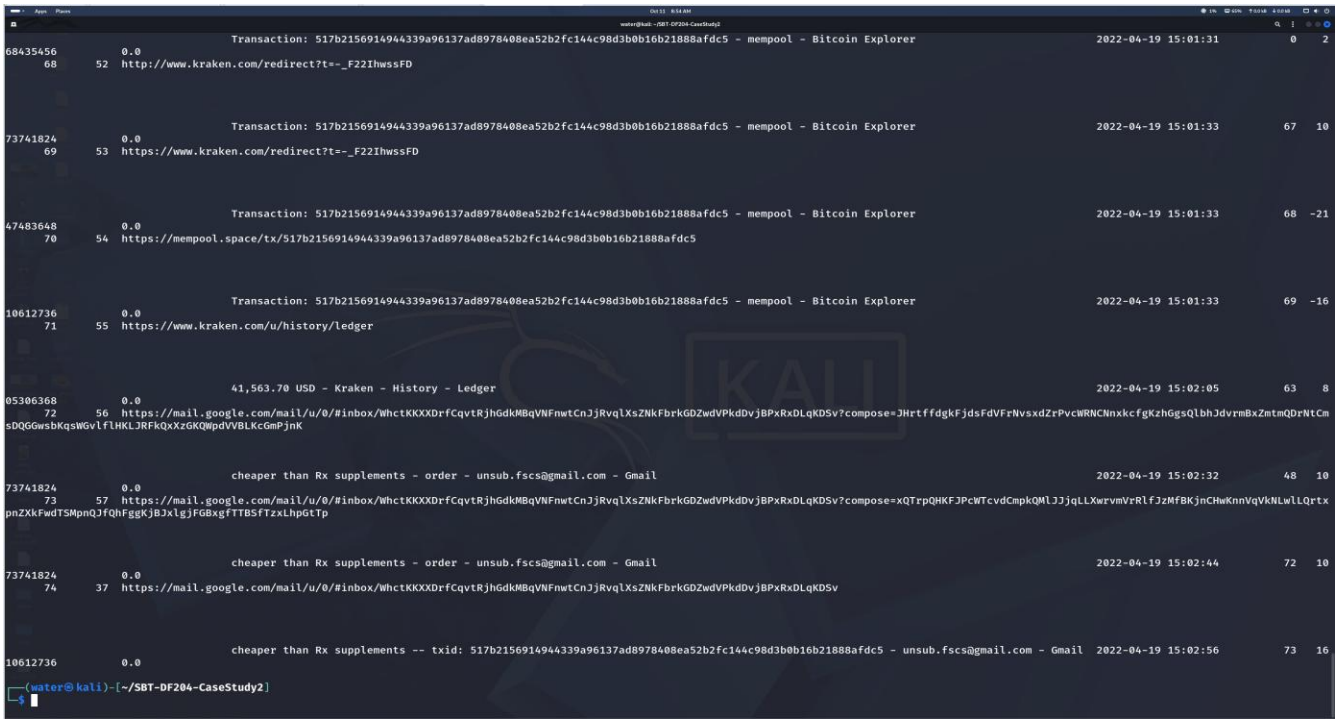
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73741824          0.0
      73      57  https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?compose=xQTrpQHKFJPcWTcvdCmpkQMlJJjqLLXwrvmVrRlfJzMfBKjnCHwKnnVqVkNLwLLQrtxpnZXkFwdTSMpnQJfQhFggKjBJxlgjFGBxgfTTBSfTzxLhpGtTp



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10612736          0.0

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
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Viewing the recorded URLs

SELECT id, url, title

FROM urls

ORDER BY id;

```
sqlite> SELECT id, url, title
FROM urls
ORDER BY id;
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id	url	title
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2	https://www.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
3	https://geo.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
4	https://baltimore.craigslist.org/	craigslist: baltimore jobs, apartments, for sale, services, community, and events
5	https://accounts.craigslist.org/login/home	craigslist account
6	https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L	craigslist - account log in
7	https://post.craigslist.org/c/bal	baltimore | choose type
8	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5	baltimore | posting confirmation
9	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type	baltimore | choose type
10	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat	baltimore | choose category
11	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit	baltimore | create posting
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13	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage	baltimore | create posting
14	https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview	baltimore | create posting
15	https://post.craigslist.org/manage/7473121658	baltimore | manage posting
16	https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html	cheaper than Rx supplements - health and beauty - by owner -...
17	https://www.google.com/search?q=gmail&rlz=1C1ONGR_enUS989US989&oq=gmail&aqs=chrome..69i57j0i433i512l6j0i131i433i512j0i433i512.1553j0j4&sourceid=chrome&ie=UTF-8	gmail - Google Search
18	https://www.google.com/gmail/	Gmail: Free, Private & Secure Email | Google Workspace
19	https://mail.google.com/mail/	Inbox (12) - unsub.fscs@gmail.com - Gmail
20	https://mail.google.com/mail/u/0/	Inbox (12) - unsub.fscs@gmail.com - Gmail
21	https://accounts.google.com/ServiceLogin?service=mail&passive=1209600&osid=1&continue=https://mail.google.com/mail/u/0/&followup=https://mail.google.com/mail/u/0/&emr=1	Gmail: Free, Private & Secure Email | Google Workspace

```
water@kali: ~/SBT-DF204-CaseStudy2
```

22	https://mail.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace
23	https://www.google.com/intl/en/mail/help/about.html	Gmail: Free, Private & Secure Email | Google Workspace
24	https://www.google.com/gmail/about/	Gmail: Free, Private & Secure Email | Google Workspace
25	https://accounts.google.com/AccountChooser/signinchooser?service=mail&continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&flowName=GlifWebSignIn&flowEntry=AccountChooser	Gmail
26	https://accounts.google.com/AccountChooser?service=mail&continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&flowName=GlifWebSignIn&flowEntry=AccountChooser	Gmail
27	https://accounts.google.com/ServiceLogin?continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&service=mail&sacu=1&rip=1	Gmail
28	https://accounts.google.com/signin/v2/identifier?continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&service=mail&sacu=1&rip=1&flowName=GlifWebSignIn&flowEntry=ServiceLogin	Gmail
29	https://accounts.google.com/signin/v2/challenge/pwd?continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&service=mail&sacu=1&rip=1&flowName=GlifWebSignIn&flowEntry=ServiceLogin&cid=1&navigationDirection=forward&TL=AM3QAYaTBRT2HiBnwAVfbyUl6WFCiAcnO3Di5PInPClfYKBQPh15Z9040uek44c3	Gmail
30	https://accounts.google.com/CheckCookie?continue=https%3A%2F%2Fmail.google.com%2Fmail%2F&service=mail&hl=en&checkedDomains=youtube&checkConnection=youtube%3A188%3A0&pstMsg=1&ifkv=AU9NCcxu9S3lHlJBnqjT5qedCC0vpqV4xmOCTcO3oAhh0gszB2CF_ogO45ADbccjsTMkTpRelEs45A&chtml=LoginDoneHtml&gidl=EgIIAA	Inbox (12) - unsub.fscs@gmail.com - Gmail
31	https://mail.google.com/accounts/SetOSID?authuser=0&continue=https%3A%2F%2Faccounts.youtube.com%2Faccounts%2FSetSID%3Fssdc%3D1%26sidt%3DALWU2ctdRyPZH5SUntnYwGNdrlnbR2cZaTgK6r%252FbZ8qgPbPyFCbHk%252F%252Fimf7iJgw0hkQX4U95uAr373wIkMusnsCUpbQjMKqit%252BdSdlrp26pauGLW7jIRZLQNIXLnHES2aA7rA71zOXb2GTTJXNJd...	Inbox (12) - unsub.fscs@gmail.com - Gmail
32	https://accounts.youtube.com/accounts/SetSID?ssdc=1&sidt=ALWU2ctdRyPZH5SUntnYwGNdrlnbR2cZaTgK6r/bZ8qgPbPyFCbHk//imf7iJgw0hkQX4U95uAr373wIkMusnsCUpbQjMKqit%2BdSdlrp26pauGLW7jIRZLQNIXLnHES2aA7rA71zOXb2GTTJXNJdeNVKWd9C5SEPlsrebTkk5rEdQJPGaS9nlqhKczzb/2kdVhK633u4Im/o9n/u0DaurpLRAdzuafm5RWTo1bkUSoZuNuchw...	Inbox (12) - unsub.fscs@gmail.com - Gmail
33	https://mail.google.com/mail/u/0/#inbox	Inbox (12) - unsub.fscs@gmail.com - Gmail
34	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXKdXnJTmxwZwPtjVmvHMrfPPRthPXJgXngHcFLmggWv	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
35	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXKdXnJTmxwZwPtjVmvHMrfPPRthPXJgXngHcFLmggWv?compose=JHrtffdgkFjdrvzMfJPRMHgQmSZBjzPlcXtWPRlXfSGwhzrQgskTqNxQLFWSmhvscKrDpXcPbHKpCDnqPgtmpWSJWbLlCLcpRBwKDFNRmPLzKkvkC	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
36	https://mail.google.com/mail/u/0/#inbox/WhctKKXXDrfCqXBGCnHkHGxZnZbkXKdXnJTmxwZwPtjVmvHMrfPPRthPXJgXngHcFLmggWv?	cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail



```
   WhctKKXXDrfCqXBGCnHkHGxZnZbkXKdXnJTmxwZwPtjVmvHMrfPPRthPXJgXngHcFLmggWv?
   compose=DXDwSWwxqgzWzDRtRJhRTXQgRWQdlXLDsZNVCgswmTqsxJBwBsdXxSKvPCxdbcBxxgBd
   wWDSvwScdmrjTqtTPCDbWhdHHQqGsTtBblLnfbhKchZLqdPKdpwL

37 https://mail.google.com/mail/u/0/#inbox/                          cheaper than Rx supplements -- txid: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - unsub.fscs@gmail.com
   WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv      - Gmail

38 https://www.google.com/url?q=https://imgur.com/dTgrkP7&source=gmail&ust=     Imgur: The magic of the Internet
   1650466604648000&usg=AOvVaw0058wuJx6Jub4I72ZRaQpM

39 https://imgur.com/dTgrkP7                                         Imgur: The magic of the Internet

40 https://www.google.com/search?q=blockchain+explorer&rlz=1C1ONGR_enUS989US989 blockchain explorer - Google Search
   &oq=blockchain+explorer&aqs=chrome..69i57j35i39j0i512l8.3445j0j7&sourceid=
   chrome&ie=UTF-8

41 https://www.blockchain.com/explorer                                Blockchain Explorer - Search the Blockchain | BTC | ETH | BCH

42 https://www.blockchain.com/search?search=                          Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer
   517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5

43 https://www.blockchain.com/btc/tx/517b2156914944339a96137ad8978408ea52b2fc14 Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 | Blockchain Explorer
   4c98d3b0b16b21888afdc5

44 https://www.blockchain.com/btc/address/38RcsURWCDmCbYocmsZCnGz1FtD7x477mt    Address: 38RcsURWCDmCbYocmsZCnGz1FtD7x477mt | Blockchain Explorer

45 https://www.google.com/search?q=kraken&rlz=1C1ONGR_enUS989US989&oq=kraken&   kraken - Google Search
   aqs=chrome..69i57j46i67i433j0i67j0i67i433j0i67j0i67i433j0i67j0i131i433i512j0
   i512l2.2504j0j7&sourceid=chrome&ie=UTF-8

46 https://www.kraken.com/en-us/sign-in                              Sign in to Kraken - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)

47 https://www.kraken.com/en-us/device-approval                      Approve new device - Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)

48 https://www.kraken.com/u/trade?i=true

49 https://www.kraken.com/u/instant                                   Kraken | Buy, Sell and Margin Trade Bitcoin (BTC) and Ethereum (ETH)

50 https://www.kraken.com/u/funding                                  41,543.80 USD - Kraken - Funding - Overview

51 https://www.kraken.com/redirect?url=https%3A%2F%2Fmempool.space%2Ftx%        Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
   2F517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5

52 http://www.kraken.com/redirect?t=-_F22IhwssFD                     Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer

53 https://www.kraken.com/redirect?t=-_F22IhwssFD                    Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer

54 https://mempool.space/tx/517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b Transaction: 517b2156914944339a96137ad8978408ea52b2fc144c98d3b0b16b21888afdc5 - mempool - Bitcoin Explorer
   16b21888afdc5

55 https://www.kraken.com/u/history/ledger                            41,563.70 USD - Kraken - History - Ledger

56 https://mail.google.com/mail/u/0/#inbox/                          cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
   WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?
   compose=JHrtffdgkFjdsFdVFrNvsxdZrPvcWRNCNnxkcfgKzhGgsQlbhJdvrmBxZmtmQDrNtCms
```

```
56 https://mail.google.com/mail/u/0/#inbox/                          cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
   WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?
   compose=JHrtffdgkFjdsFdVFrNvsxdZrPvcWRNCNnxkcfgKzhGgsQlbhJdvrmBxZmtmQDrNtCms
   DQGGwsbKqsWGvlflHKLJRFkQxXzGKQWpdVVBLKcGmPjnK

57 https://mail.google.com/mail/u/0/#inbox/                          cheaper than Rx supplements - order - unsub.fscs@gmail.com - Gmail
   WhctKKXXDrfCqvtRjhGdkMBqVNFnwtCnJjRvqlXsZNkFbrkGDZwdVPkdDvjBPxRxDLqKDSv?
   compose=xQTrpQHKFJPcWTcvdCmpkQMlJJjqLLXwrvmVrRlfJzMfBKjnCHwKnnVqVkNLwlLQrtxp
   nZXkFwdTSMpnQJfQhFggKjBJxlgjFGBxgfTTBSfTzxLhpGtTp
sqlite>
```

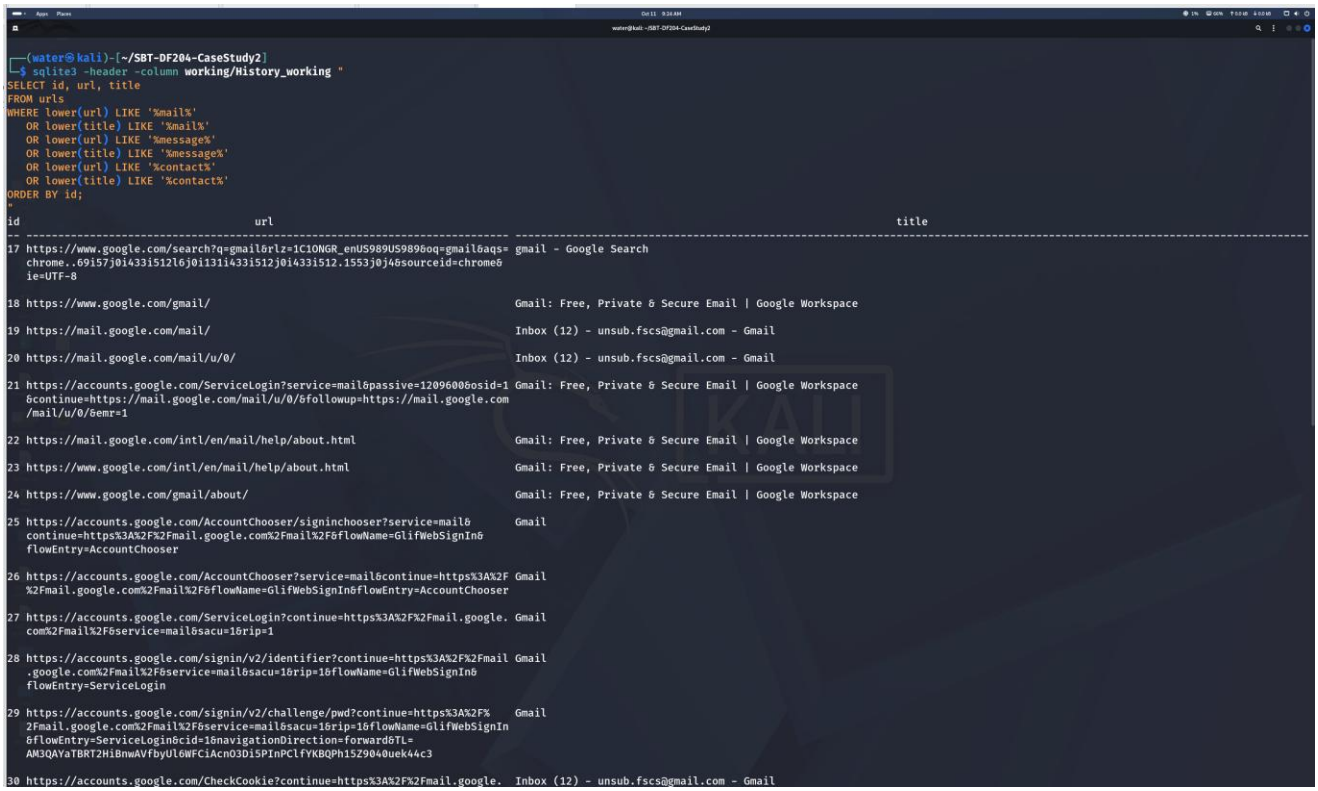
Checking history for relevant activity on the file

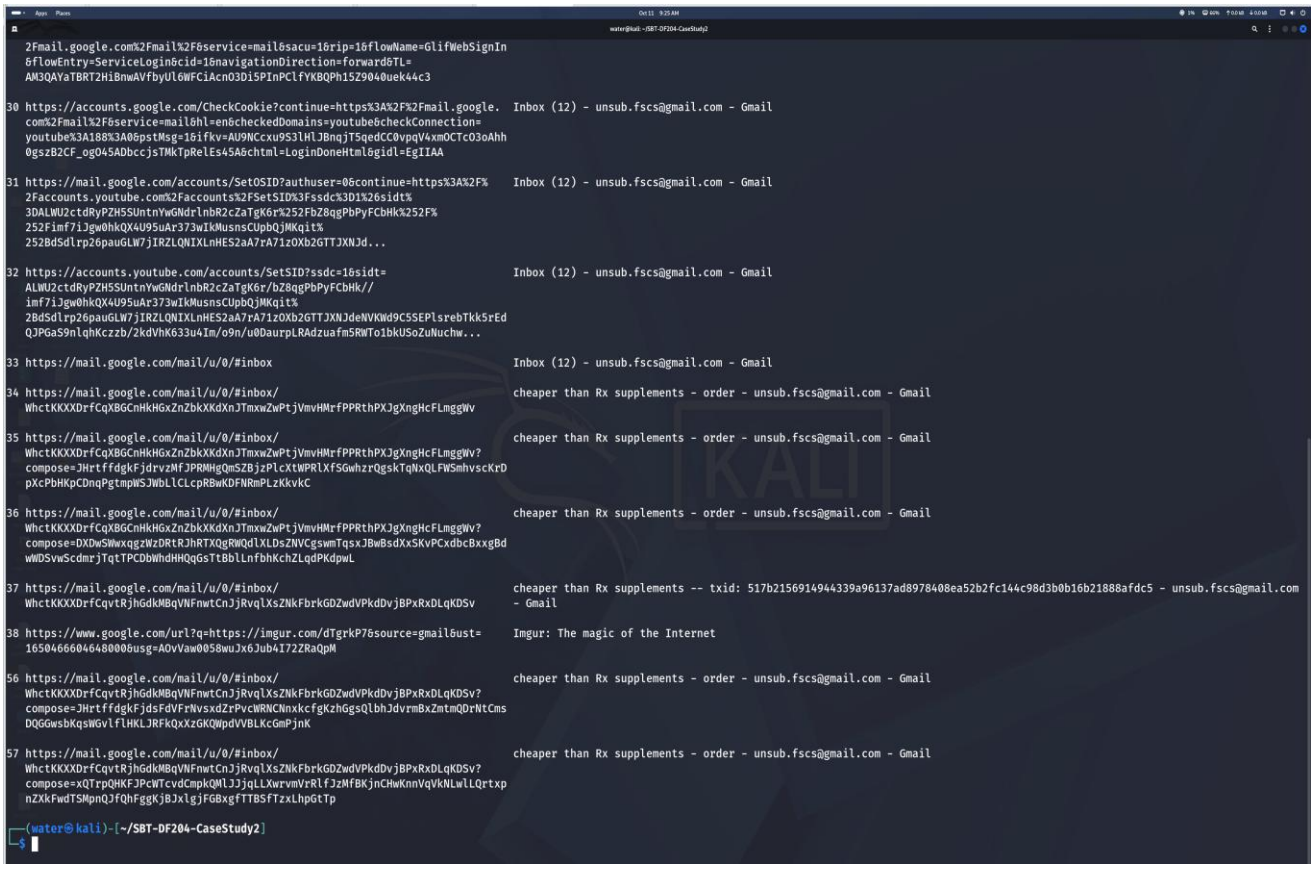
```
sqlite3 -header -column working/History_working "
SELECT id, url, title
FROM urls
WHERE lower(url) LIKE '%mail%'
   OR lower(title) LIKE '%mail%'
   OR lower(url) LIKE '%drug%'
   OR lower(title) LIKE '%drug%'
   OR lower(url) LIKE '%crypto%'
   OR lower(title) LIKE '%crypto%'
   OR lower(url) LIKE '%bitcoin%'
   OR lower(title) LIKE '%bitcoin%'
   OR lower(url) LIKE '%payment%'
   OR lower(title) LIKE '%payment%'
ORDER BY id;
"
```

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working "
SELECT id, url, title
FROM urls
WHERE lower(url) LIKE '%mail%'
   OR lower(title) LIKE '%mail%'
   OR lower(url) LIKE '%drug%'
   OR lower(title) LIKE '%drug%'
   OR lower(url) LIKE '%crypto%'
   OR lower(title) LIKE '%crypto%'
   OR lower(url) LIKE '%bitcoin%'
   OR lower(title) LIKE '%bitcoin%'
   OR lower(url) LIKE '%payment%'
   OR lower(title) LIKE '%payment%'
ORDER BY id;
"
id                                      url                                                
```

Checking for Communication related records

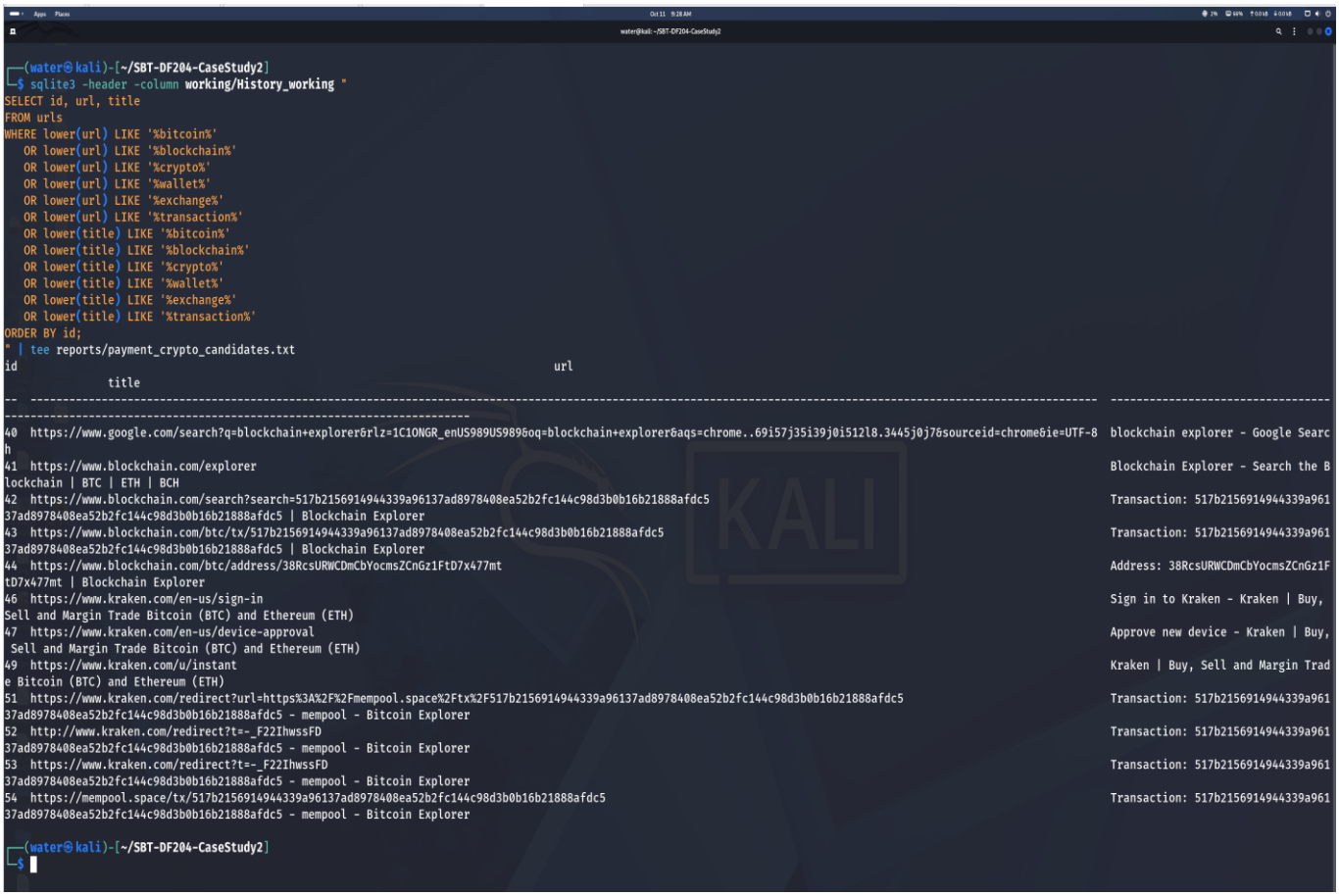
sqlite3 -header -column working/History_working "
SELECT id, url, title
FROM urls
WHERE lower(url) LIKE '%mail%'
  OR lower(title) LIKE '%mail%'
  OR lower(url) LIKE '%message%'
  OR lower(title) LIKE '%message%'
  OR lower(url) LIKE '%contact%'
  OR lower(title) LIKE '%contact%'
ORDER BY id;
"





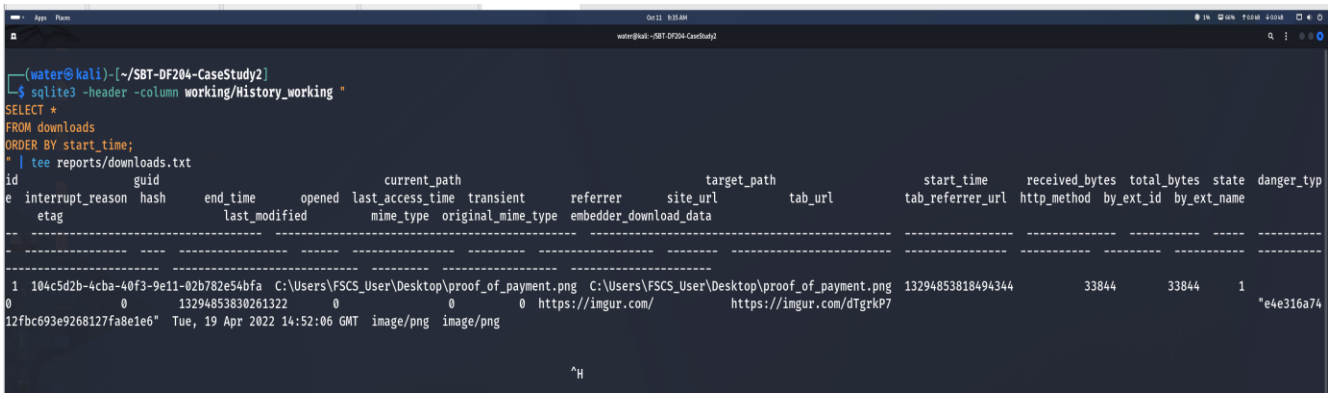
Checking for cryptocurrency/payment activity

```
sqlite3 -header -column working/History_working "
SELECT id, url, title
FROM urls
WHERE lower(url) LIKE '%bitcoin%'
   OR lower(url) LIKE '%blockchain%'
   OR lower(url) LIKE '%crypto%'
   OR lower(url) LIKE '%wallet%'
   OR lower(url) LIKE '%exchange%'
   OR lower(url) LIKE '%transaction%'
   OR lower(title) LIKE '%bitcoin%'
   OR lower(title) LIKE '%blockchain%'
   OR lower(title) LIKE '%crypto%'
   OR lower(title) LIKE '%wallet%'
   OR lower(title) LIKE '%exchange%'
   OR lower(title) LIKE '%transaction%'
ORDER BY id;
" | tee reports/payment_crypto_candidates.txt
```



Checking for Downloads and Download URL chain

sqlite3 -header -column working/History_working "
SELECT *
FROM downloads
ORDER BY start_time;
" | tee reports/downloads.txt



Download URL Chain

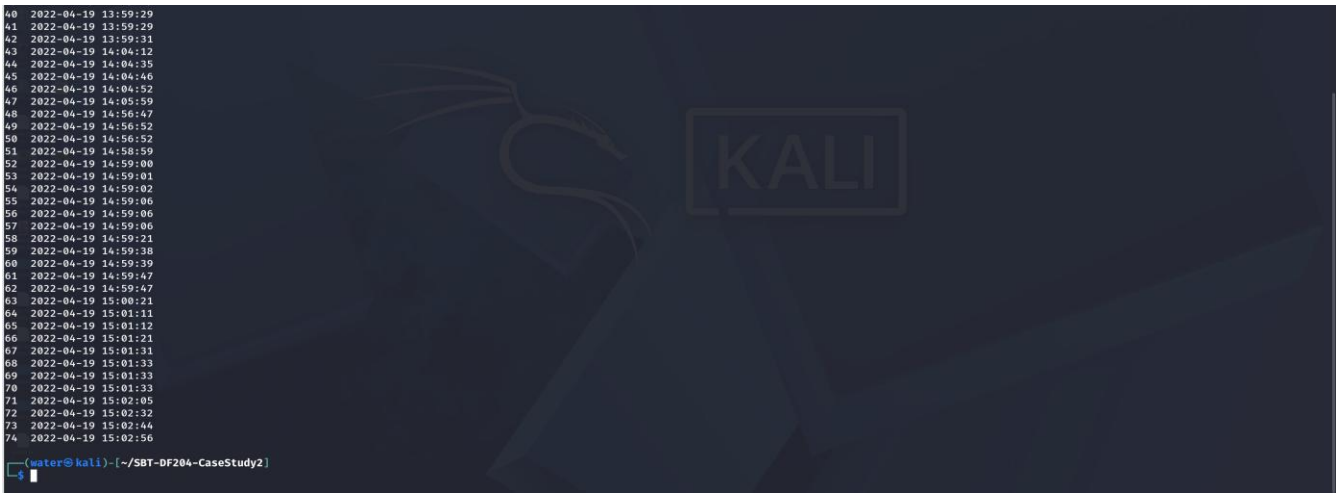
```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working "
SELECT *
FROM downloads_url_chains
ORDER BY id;
" | tee reports/download_url_chains.txt
id  chain_index  url
--  -----------  ------------------------------
1           0  https://i.imgur.com/dTgrkP7.png

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
```

Converting Timestamps

```
sqlite3 -header -column working/History_working "
SELECT
    id,
    datetime(
        visit_time/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS visit_utc
FROM visits
ORDER BY visit_time;
"
```

```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working "
SELECT
    id,
    datetime(
        visit_time/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS visit_utc
FROM visits
ORDER BY visit_time;
"
id       visit_utc
--  -------------------
1   2022-04-19 13:54:18
2   2022-04-19 13:54:19
3   2022-04-19 13:54:19
4   2022-04-19 13:54:19
5   2022-04-19 13:54:19
6   2022-04-19 13:54:22
7   2022-04-19 13:54:22
8   2022-04-19 13:54:39
9   2022-04-19 13:55:03
10  2022-04-19 13:55:03
11  2022-04-19 13:55:05
12  2022-04-19 13:55:05
13  2022-04-19 13:55:05
14  2022-04-19 13:55:10
15  2022-04-19 13:55:15
16  2022-04-19 13:58:13
17  2022-04-19 13:58:17
18  2022-04-19 13:58:19
19  2022-04-19 13:58:22
20  2022-04-19 13:58:27
21  2022-04-19 13:58:31
22  2022-04-19 13:58:38
23  2022-04-19 13:58:39
24  2022-04-19 13:58:40
25  2022-04-19 13:58:40
26  2022-04-19 13:58:40
27  2022-04-19 13:58:40
28  2022-04-19 13:58:40
29  2022-04-19 13:58:40
30  2022-04-19 13:58:40
31  2022-04-19 13:58:43
32  2022-04-19 13:58:43
33  2022-04-19 13:58:43
34  2022-04-19 13:58:44
35  2022-04-19 13:58:44
36  2022-04-19 13:59:26
37  2022-04-19 13:59:29
38  2022-04-19 13:59:29
39  2022-04-19 13:59:29
40  2022-04-19 13:59:29
```



Identifying the earliest and latest recorded activity

```
sqlite3 -header -column working/History_working "
SELECT
    datetime(
        MIN(visit_time)/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS earliest_visit_utc,
    datetime(
        MAX(visit_time)/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS latest_visit_utc,
    COUNT(*) AS total_visits
FROM visits;
" | tee reports/visit_time_range.txt
```

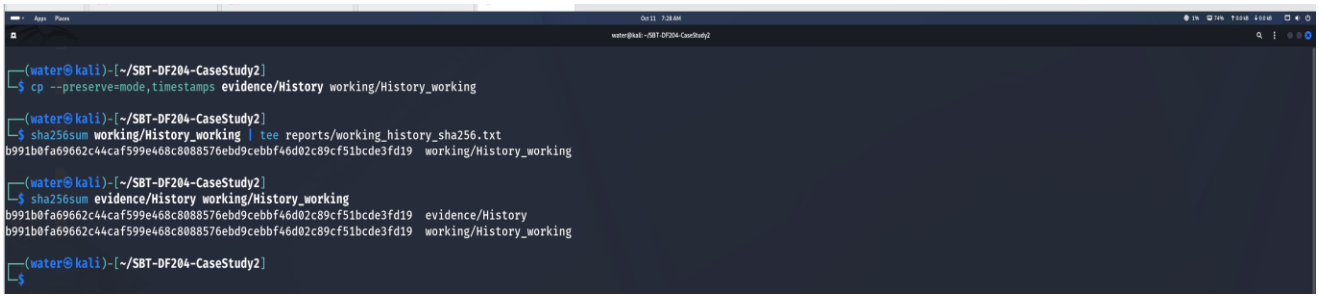


Below is the analysis and result to the eight questions asked on the case study and also relating to all the screenshot above

- **What evidence file did you acquire and how did you preserve it?,** Report the original file name, source, file size, SHA-256 value, working-copy method, and any file-access precautions.

The evidence acquired was a Google Chrome History SQLite database named History, supplied as part of the Case Study 2 training materials. The original file was stored in the evidence/History folder on Kali Linux, and its SHA-256 hash was recorded as b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19. A working copy was created at working/History_working using the command cp --preserve=mode,timestamps evidence/History working/History_working. The original and working-copy hashes were compared and found to be identical, confirming that the copy matched the original at the time of verification. The stat command was used to record the original file's name, size, permissions, and owner. The original evidence was kept separate from the working copy, and SQLite queries were performed on the working copy to reduce the risk of changing the original evidence.

sha256sum working/History_working | tee reports/working_history_sha256.txt
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History_working
sha256sum evidence/History working/History_working
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 evidence/History
b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 working/History_working





- **What browser-history tables and fields are relevant?** Identify the tables you examined and explain the role of the urls, visits, downloads, and downloads_url_chains data used in your case.

The Chrome History database was examined using SQLite commands on Kali Linux to identify records related to browser activities. Four main tables were examined: urls, visits, downloads, and downloads_url_chains. The urls table contains website addresses and page titles, which help identify websites that may be related to the suspected activities. The visits table records browsing visits, timestamps, navigation transitions, and visit duration, helping to establish the order of events. The downloads and downloads_url_chains tables provide information about downloaded files and their source URLs, helping to investigate whether any downloads may be connected to the case.

Table	**Relevant fields**	**Purpose in the investigation**
urls	id, url, title	Identifies visited website addresses and page titles that may relate to advertising, communication, or cryptocurrency payments.
visits	id, url, visit_time, from_visit, transition, visit_duration	Shows when visits occurred, links visits to URL records, and helps reconstruct browsing order and navigation patterns.
downloads	start_time and other available download fields	Helps identify downloaded files and examine their recorded details and timestamps.
downloads_url_chains	Available URL-chain fields, such as download identifiers and source URLs where present	Helps trace the URLs associated with a download and understand where the downloaded file originated.

- **When did the recorded web activity occur?** Convert Chrome WebKit timestamps to UTC and provide the query or method. State any local-time conversion separately.

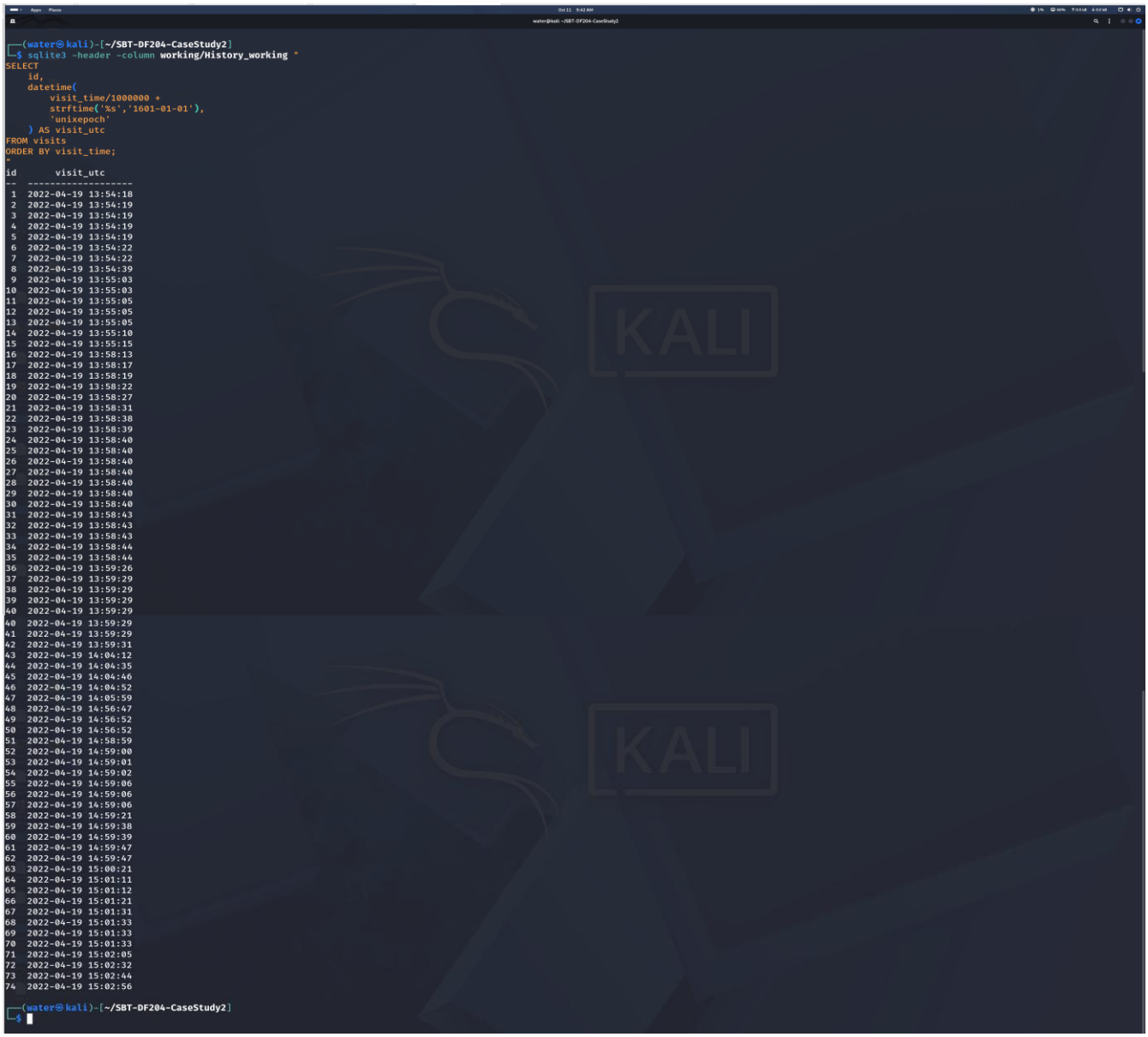
The recorded web activity was examined using the Google Chrome History SQLite database on Kali Linux. Chrome stores browser visit timestamps as WebKit timestamps, which represent the number of microseconds since 1 January 1601 (UTC). The visits table was queried using SQLite

to convert these timestamps into readable UTC dates and times. The converted timestamps were arranged in ascending order to establish the sequence of browsing activities.
sqlite3 -header -column working/History_working " SELECT id, datetime(visit_time/1000000 + strftime('%s','1601-01-01'), 'unixepoch') AS visit_utc FROM visits ORDER BY visit_time; " | tee reports/visit_timestamps_utc.txt

visit_time contains the Chrome WebKit timestamp in microseconds.
Dividing by 1000000 converts microseconds to seconds.
strftime('%s','1601-01-01') obtains the Unix timestamp for the WebKit epoch.
datetime(..., 'unixepoch') converts the result into a readable UTC date and time.
ORDER BY visit_time arranges the visits from earliest to latest.

earliest_visit_utc latest_visit_utc total_visits
------------------ ------------------ ------------
2022-04-19 13:54:18 2022-04-19 15:02:56





```
┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$ sqlite3 -header -column working/History_working "
SELECT
    id,
    datetime(
        visit_time/1000000 +
        strftime('%s','1601-01-01'),
        'unixepoch'
    ) AS visit_utc
FROM visits
ORDER BY visit_time;
"
id       visit_utc
--  -------------------
 1  2022-04-19 13:54:18
 2  2022-04-19 13:54:19
 3  2022-04-19 13:54:19
 4  2022-04-19 13:54:19
 5  2022-04-19 13:54:19
 6  2022-04-19 13:54:22
 7  2022-04-19 13:54:22
 8  2022-04-19 13:54:39
 9  2022-04-19 13:55:03
10  2022-04-19 13:55:03
11  2022-04-19 13:55:05
12  2022-04-19 13:55:05
13  2022-04-19 13:55:05
14  2022-04-19 13:55:10
15  2022-04-19 13:55:15
16  2022-04-19 13:58:13
17  2022-04-19 13:58:17
18  2022-04-19 13:58:19
19  2022-04-19 13:58:22
20  2022-04-19 13:58:27
21  2022-04-19 13:58:31
22  2022-04-19 13:58:38
23  2022-04-19 13:58:39
24  2022-04-19 13:58:40
25  2022-04-19 13:58:40
26  2022-04-19 13:58:40
27  2022-04-19 13:58:40
28  2022-04-19 13:58:40
29  2022-04-19 13:58:40
30  2022-04-19 13:58:40
31  2022-04-19 13:58:43
32  2022-04-19 13:58:43
33  2022-04-19 13:58:43
34  2022-04-19 13:58:44
35  2022-04-19 13:58:44
36  2022-04-19 13:59:26
37  2022-04-19 13:59:29
38  2022-04-19 13:59:29
39  2022-04-19 13:59:29
40  2022-04-19 13:59:29
40  2022-04-19 13:59:29
41  2022-04-19 13:59:29
42  2022-04-19 13:59:31
43  2022-04-19 14:04:12
44  2022-04-19 14:04:35
45  2022-04-19 14:04:46
46  2022-04-19 14:04:52
47  2022-04-19 14:05:59
48  2022-04-19 14:56:47
49  2022-04-19 14:56:52
50  2022-04-19 14:56:52
51  2022-04-19 14:58:59
52  2022-04-19 14:59:00
53  2022-04-19 14:59:01
54  2022-04-19 14:59:02
55  2022-04-19 14:59:06
56  2022-04-19 14:59:06
57  2022-04-19 14:59:06
58  2022-04-19 14:59:21
59  2022-04-19 14:59:38
60  2022-04-19 14:59:39
61  2022-04-19 14:59:47
62  2022-04-19 14:59:47
63  2022-04-19 15:00:21
64  2022-04-19 15:01:11
65  2022-04-19 15:01:12
66  2022-04-19 15:01:21
67  2022-04-19 15:01:31
68  2022-04-19 15:01:33
69  2022-04-19 15:01:33
70  2022-04-19 15:01:33
71  2022-04-19 15:02:05
72  2022-04-19 15:02:32
73  2022-04-19 15:02:44
74  2022-04-19 15:02:56

┌──(water㉿kali)-[~/SBT-DF204-CaseStudy2]
└─$
```

- **How did the browser reach the material URLs?** Use transition values, transition qualifiers, and from_visit links to distinguish typed or omnibox activity, links, form submissions, redirects, and navigation chains.

The browser navigation was investigated using the Chrome History database, particularly the urls and visits tables. The transition field helps identify how a page was opened, such as through a typed address, a clicked link, or a form submission, while transition qualifiers provide additional details about the navigation. The from_visit field links a visit to an earlier visit and helps reconstruct the sequence of browser navigation. These fields can help distinguish direct navigation, links, redirects, and other navigation chains related to the material URLs.

The below command was used to get the record

```
sqlite3 -header -column working/History_working "
SELECT
   v.id AS visit_id,
   u.id AS url_id,
   u.url,
   u.title,
   datetime(
      v.visit_time/1000000 +
      strftime('%s','1601-01-01'),
      'unixepoch'
   ) AS visit_utc,
   v.from_visit,
   v.transition,
   v.visit_duration/1000.0 AS duration_ms
FROM visits AS v
JOIN urls AS u ON v.url = u.id
ORDER BY v.visit_time;
" | tee reports/full_visit_timeline.txt
```

- **What web-history evidence relates to the suspected sale or posting?** Identify the records that support or challenge an advertisement, posting workflow, management page, or related interaction. Cite the exact artefacts.

The Chrome History database was examined to identify website records that could relate to a suspected drug advertisement, posting, or sale. The urls table was used to identify website addresses and page titles, while the visits table provided timestamps and navigation details that could help establish the browsing sequence. Relevant searches included URLs and titles containing terms related to drugs, communication, email, and payments. These records can help identify possible advertising pages, posting workflows, management pages, or related interactions. However, the report provides the investigation queries but does not show the actual matching records, so no specific advertisement, posting, or sale can be confirmed from the available evidence.

Artefact	Evidence to identify	Relevance to the suspected sale or posting
working/History_working — urls table	Exact URL, page title, and URL record ID	Help identify an advertisement, posting page, or management interface.
working/History_working — visits table	Visit ID, URL ID, timestamp, transition, and from_visit	Help establish when a relevant page was visited and how the browser reached it.
reports/full_visit_timeline.txt	Chronological URL and visit records	Help determine the sequence of potentially related browsing activities.
reports/payment_crypto_candidates.txt	Matching cryptocurrency and payment URLs or titles, if any	Help identify browsing associated with payment verification or transaction services.
reports/record_counts.txt	Database record counts, if generated	Documents the number of URL, visit, and download records examined.

- **What evidence relates to communications and payment activity?** Identify relevant email, image-hosting, blockchain-explorer, exchange, transaction, wallet, or download artefacts. Explain each link without assuming content that the record does not show.

The Chrome History database was examined to identify browser records that might relate to communication and payment activities. The investigation used SQL queries to search website URLs and page titles containing terms such as mail, message, contact, Bitcoin, blockchain, cryptocurrency, wallet, exchange, and transaction. The urls table can identify relevant websites, while the visits table shows when those pages were visited and how the browsing activity was connected. The downloads and downloads_url_chains tables provide additional information about downloaded files and their recorded source URLs. However, the report does not show the actual matching records or prove that emails were sent, images were uploaded, payments were made, or cryptocurrency transactions were completed.

- **What is the reconstructed timeline?** Create a chronological table that separates direct observations from your interpretation of the activity stage.

The timeline was reconstructed using the Chrome History SQLite database on Kali Linux. The visits and urls tables were used to identify browsing times, website addresses, page titles, and navigation sequences. The recorded Chrome WebKit timestamps were converted to UTC and arranged from the earliest to the latest visit. Each event was separated into a direct observation and an interpretation of its possible activity stage. However, the exact timestamps and URLs must be taken from the actual database query results before the timeline can be considered complete.

What conclusion can you defend? Give a cautious conclusion with confidence level and at least three limitations or alternative explanations, including the difference between browser-profile activity and human identity.

- The investigation examined a Google Chrome History database using SQLite queries to identify browsing activities, timestamps, navigation records, and downloads that might relate to the suspected drug sale. However, the report does not contain the actual query results needed to confirm that an advertisement was published, communication took place, or a payment was completed. Therefore, the confidence level for linking the recorded browser activity to the suspected offence is low based on the information currently documented. One limitation is that browser history records activity associated with a browser profile but does not, by itself, identify the human who performed it. Other limitations are that some browsing records may be missing or deleted, visiting a website does not prove that a transaction occurred, and another person may have used the device or profile.

2.7. Analysis and Findings

The investigation examined a Google Chrome History database using SQLite on Kali Linux.
The urls, visits, downloads, and downloads_url_chains tables were reviewed to identify relevant browsing activities.
The recorded web activity occurred between 19 April 2022 at 13:54:18 UTC and 15:02:56 UTC.
Search queries were prepared to identify possible evidence of drug advertising, communication, cryptocurrency, payments, and downloads.
However, the actual matching records were not fully documented, so the available results do not confirm that a drug advertisement was published or that a payment was completed.

2.8. Challenges and Solutions

One challenge was understanding the structure of the Chrome History database and identifying the correct tables for the investigation. This was addressed by using SQLite commands to inspect the database tables and their fields.
Another challenge was converting Chrome timestamps into readable dates and times, which was handled using an SQL timestamp-conversion query.

2.9. Conclusion

The investigation demonstrated how Chrome browser history can be examined to reconstruct browsing activity and develop a timeline. The SHA-256 hashes of the original database and its working copy matched, supporting the integrity of the copied evidence at the time of verification. The recorded browsing period was identified as 19 April 2022, from 13:54:18 UTC to 15:02:56 UTC.
However, the documented findings do not provide enough evidence to confirm that the suspected drug sale, communication, or payment occurred. Therefore, confidence in linking the recorded browser activity to the suspected offence remains low, and further supporting evidence is required.

2.10. Recommendations

Assignment like this improved student knowledge and understanding on Forensic case giving the student the ability to think deep and do deep research before concluding on a case

2.11. References

Class materials Guide from Founder ICDFA on Basic Networking Skills for Digital Forensics, (Digital Currency)
https://onedrive.live.com/?id=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments%2FDocuments%2Fdigital%2Dforensics%2Dlab%2FDigital%5FCurrency&listurl=%2Fpersonal%2Fa674908a5e2e2f2a%2FDocuments&ithint=folder&e=sycXyz&migratedtospo=true&redeem=aHR0cHM6Ly8xZHJ2Lm1zL2YvYy9hNjc0OTA4YTVlMmUyZjJhL0lnRDJXbjdIeEhfMlRJVGVDaERWcE9yZUFmQ3BPQlNFaExpb1dYcGNySFZBVmlzP2U9c3ljWHl6&ga=1